

Autonomous Surface Robot for Depth-Resolved Environmental Data Collection

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Abstract - This paper presents the design and deployment of an autonomous surface vehicle (ASV) equipped with a motorized winch payload to measure the attenuation of sunlight with respect to depth in Dana Point Harbor, serving as an indicator of water clarity and suspended pollution. To collect stable data, the robot executed a stepped descent, lowering the sensor payload containing an amorphous silicon solar cell (light), an NTC thermistor (temperature), and a pressure sensor (depth). During the field deployment, the system successfully lowered the payload to a maximum depth of 1.84 meters. The collected data revealed an exponential decay in light intensity, dropping from an estimated 100,000 Lux at the surface to approximately 88,000 Lux at maximum depth. Although the thermistor experienced a hardware short that railed out its output, the successful light and pressure readings accurately mapped the harbor's light attenuation. Ultimately, these results prove that low-cost ASVs with stepped descent winches are highly effective for depth-resolved environmental monitoring, provided that the underwater sensor has ample waterproofing.

I. INTRODUCTION

Coastal ecosystems and harbor environments are highly sensitive to runoff which can significantly alter water quality and marine habitats. The presence of total suspended sediments (TSS), colored dissolved organic matter (CDOM), and phytoplankton directly impact the optical properties of coastal waters which are the primary causes of light attenuation [1]. The contributions of these factors to overall light attenuation in coastal embayment can be modeled mathematically and visualized (see Figure 1 [1]). Monitoring these variables is necessary since water monitoring sensors enable the collection of data necessary for evaluating ecosystem health [2]. Measuring how light penetrates the water serves as a highly effective way for assessing water clarity and the concentration of suspended pollution in harbors like Dana Point.

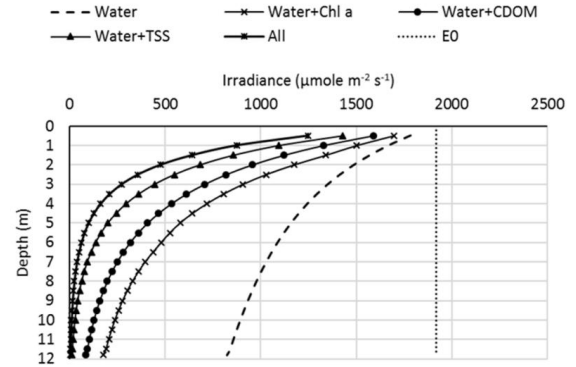


Figure 1: Quantifying contributions to light attenuation in coastal embayments [1]

This light attenuation primarily occurs in the epipelagic zone, the uppermost layer of the ocean extending from the surface down to 200 meters [3]. In this sunlight zone, light intensity decreases exponentially with depth as electromagnetic radiation is absorbed and scattered by water molecules and suspended matter [4]. Additionally, the distribution of the sun's heat creates thermal gradients within this top layer [3]. Local surface temperatures in Dana Point vary seasonally but generally provide a baseline for expected thermal gradients [5]. By simultaneously collecting both light intensity and water temperature, researchers can identify thermal gradients and locate suspended pollutants introduced by shoreline runoff.

To accurately capture these depth-resolved measurements, autonomous surface vehicles (ASVs) offer reliable capabilities that improve traditional environmental monitoring [6]. For this study, an ASV was designed and deployed to autonomously navigate coastal waters and deliver a specialized sensor payload. The vehicle utilized a motorized winch system programmed with a stepped descent, allowing light, temperature, and pressure sensors to stabilize and record data at discrete depth intervals. This system provided a controlled methodology for gathering data without requiring a fully submersible vehicle.

The primary goal of this project was to measure the attenuation of sunlight with respect to depth in Dana Point Harbor to evaluate local water clarity and the optical impact of suspended harbor pollution. A secondary goal was to correlate these light profiles with temperature gradients to identify potential mixing in the shallow water. The project also aimed to demonstrate the mechanical and electrical viability of a low-cost, automated winch system mounted on a surface robot for reliable, high-resolution environmental data collection.

II. SENSOR DESIGN

A. Pressure Sensor (Depth)

Mapping temperature and light intensity gradients requires accurate vertical positioning within the water. Therefore, a pressure sensor was selected to calculate the vehicle's payload depth. The MPX5700 pressure sensor was chosen for this application because it provides a voltage output that scales linearly with absolute pressure [7].

The MPX5700 operates over a pressure range of 0 to 700 kPa [7]. The expected deployment in Dana Point Harbor required traversing from the surface down to roughly 2 meters. Standard atmospheric pressure at sea level is approximately 101 kPa, and seawater pressure increases by roughly 9.8 kPa per meter of depth so the maximum expected pressure during deployment was approximately 126 kPa. This environmental excitation range (101 kPa to 126 kPa) utilizes only 18% of the sensor's full-scale span and falls well within the datasheet's maximum operational limits [7].

Powered by a 5.0V supply, the sensor datasheet specifies a typical raw output range of (0.2V to 4.7V) [7]. The sensor's raw voltage equation is defined in (Eq.1).

$$V_{sens} = V_s \cdot ((0.0012858 \cdot P) + 0.04) \quad (\text{Eq.1})$$

Where P is pressure in kPa [7]. Based on this equation, the raw sensor output at the surface (101kPa) is approximately 0.85V, and the raw output at the maximum deployment depth (126kPa) is approximately 1.01V. This narrow 0.16 V range is insufficient to utilize the 0 to 3.3V analog-to-digital converter (ADC) resolution on the Teensy microcontroller.

To map this narrow output span into a wider, measurable voltage range, an inverting operational amplifier circuit was designed. As shown in (Figure 2), the raw sensor output passes through a 10 kΩ input resistor and a 115 kΩ feedback resistor, yielding a gain of -11.5. A voltage divider biases the non-inverting input, adding a fixed positive DC offset. The resulting voltage equation is (Eq.2)

$$V_{circuit} = 12.5 - (11.5 \cdot V_{sens}) \quad (\text{Eq.2})$$

This signal successfully maps the 0.85V surface reading to approximately 2.7 V at the ADC, and the 1.01V maximum depth reading to approximately 1.0V at the ADC.

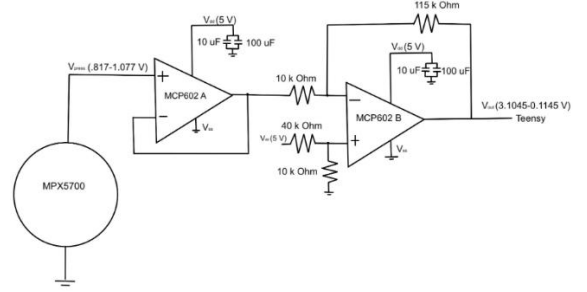


Figure 2: Pressure Sensor schematic

To confirm that the sensor and the circuit worked as intended, the payload was submerged to known depths, and the output voltages were recorded. The resulting calibration curve (See Figure 3) demonstrated a high linear relationship ($R^2 = 0.999$) modeled by the equation (Eq.3)

$$\text{Depth} = (0.814 \cdot V_{circuit}) - 0.0381 \quad (\text{Eq.3})$$

Software modifications were implemented on the Teensy to execute (Eq.3) in real time. Because the mechanical winch lowers the payload at a slow, rate, the 10 Hz sampling rate of the data logging software was sufficient to capture the depth data without aliasing.

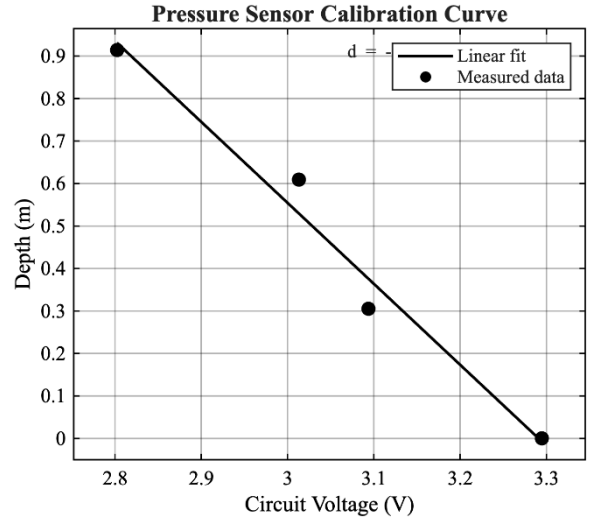


Figure 3: Calibration curve correlating the voltage output to water depth.

B. Solar Panel Sensor (Light Intensity)

To measure the attenuation of sunlight through the water, the Panasonic AM-5412CAR-DGK-T amorphous silicon solar cell was selected [8]. Amorphous silicon cells are highly sensitive to the visible light spectrum, specifically the wavelengths that penetrate shallow seawater and provide a large

surface area for gathering light in murky aquatic environments [9].

The expected environmental excitation ranges from complete darkness (0 Lux) to maximum deployment depth or under heavy overcast conditions, up to full direct summer sunlight at the water's surface (approximately 100,000 Lux). The AM-5412CAR-DGK-T is designed to operate across this range making it suitable for varying coastal sunlight conditions throughout the day [9].

As shown in the circuit schematic (See Figure 4), the positive terminal of the solar panel was wired directly to the Teensy's analog input pin (A0), and the negative terminal was wired directly to the common ground. Because the Teensy's analog-to-digital converter (ADC) has a high input impedance, the circuit draws negligible current from the panel. Therefore, the sensor operates in an unloaded state, and the microcontroller directly measures the panel's open circuit voltage V_{open} .

According to the manufacturer's specifications, the amorphous silicon cell outputs a maximum open circuit voltage of approximately 3.3V to 3.4V under high illumination [9]. This matches the 0 to 3.3V input limits of the Teensy ADC, ensuring the sensor utilizes the full range of the microcontroller without requiring voltage dividers or op-amp interface circuits.

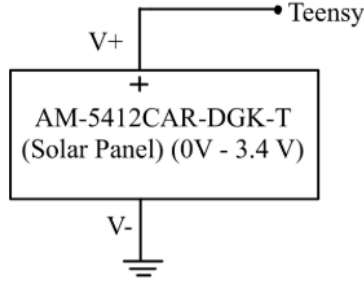


Figure 4: Circuit schematic for the AM-5412CAR-DGK-T solar sensor.

Since the open circuit voltage of a photovoltaic cell does not scale linearly with light intensity, it scales logarithmically. To map the measured voltage back to a linear Lux value for environmental data collection, an exponential calibration function was implemented in the Teensy software (Eq.4)

$$Lux = A \cdot (e^{B \cdot V_{open}} - 1) \quad (\text{Eq.4})$$

By modifying the constants until the Lux value matched known ambient light in a lab room setting with a Lux meter, the calibration yielded optimal constants $A = 10$ and $B = 2.79$. To prevent noise from generating false light readings in near

dark conditions, a limit was introduced to output 0 Lux for any voltage below 0.05 V. Additionally, the software clamped the maximum output to 100,000 Lux to represent surface saturation.

The sensor's performance and calibration were verified during the harbor deployment. As seen in (Figure 5), the calculated Lux versus Depth plot demonstrates a clear, continuous decay in light intensity as the payload descended, matching the physical expectations. (Figure 6) isolates a specific segment of the deployment from (58 to 70 seconds), to make the relationship clearer. Given the slow mechanical descent rate of the winch system, the standard 10 Hz sampling rate of the data logger was sufficient to capture the light data without aliasing.

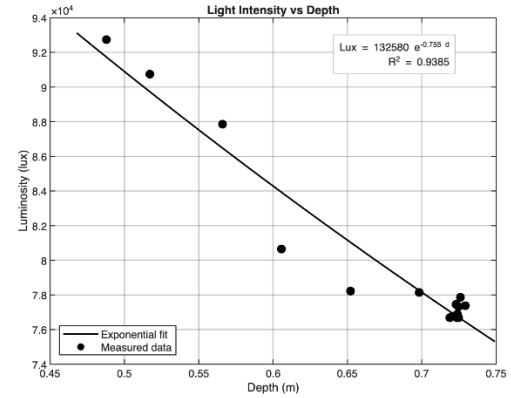


Figure 5: Calculated light intensity (Lux) versus depth (meters) during deployment

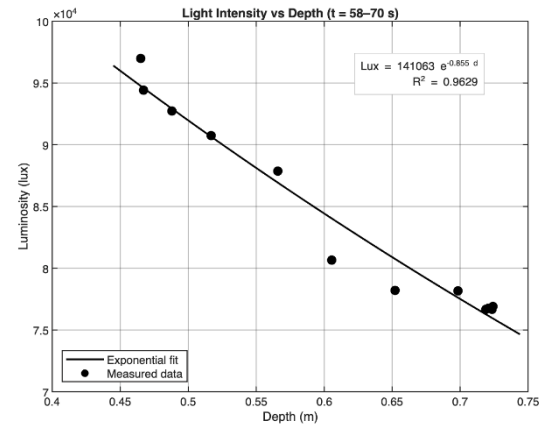


Figure 6: Light intensity data between 58 and 70 seconds

C. Thermistor (Water Temperature)

Mapping the thermal gradient of the epipelagic zone requires a sensor capable of detecting minor temperature variations with a rapid response time. The expected coastal water temperatures in Dana

Point typically range from 14°C to 18°C during the Spring. To capture this temperature, an operational input range of 10°C to 20°C was established. The Murata NXFT15WB473FA2B150 Negative Temperature Coefficient (NTC) thermistor was selected to make this measurement [10]. This specific sensor features a small resin-coated head that provides a fast thermal time constant and high sensitivity, making it ideal for detecting temperatures as the payload descends through the water [10].

To interface the variable resistance of the NTC thermistor with the microcontroller, a voltage divider circuit was designed (See Figure 7). The thermistor was connected to the 5V supply, while a 100 kΩ fixed reference resistor was connected to ground. The analog output voltage V_{out} was measured at the junction between the two components. Based on the datasheet, the thermistor's resistance decreases non-linearly as temperature increases [10]. However, over the narrow 10°C to 20°C deployment range, the sensor behavior can be closely approximated by a linear model.

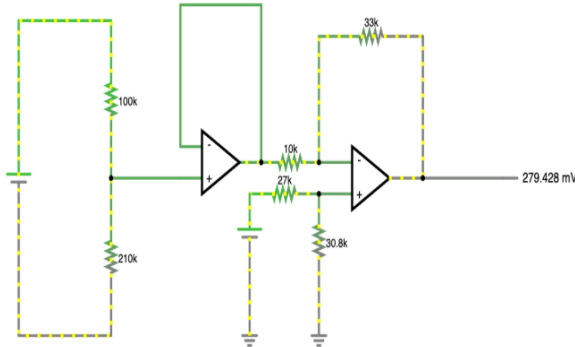


Figure 7: Voltage divider schematic for the Murata NTC thermistor

Calibration of the thermistor generated a linear equation relating resistance to temperature seen in (Eq.5)

$$R = (-6.95 \cdot T) + 259 \quad (\text{Eq.5})$$

with an ($R^2 = 0.997$) as shown in (Figure 8). Using this calibration model, the expected resistance at 10°C is approximately 189.5 kΩ, and the resistance at 20°C is roughly 120 kΩ. Passing these resistance limits through the voltage divider equation (Eq.6)

$$V_{out} = 5 \cdot \frac{100}{100 + R_{th}} \quad (\text{Eq.6})$$

which yields an expected output voltage range of 1.14V to 1.50V which is in the 0 to 3.3V bounds of the Teensy ADC.

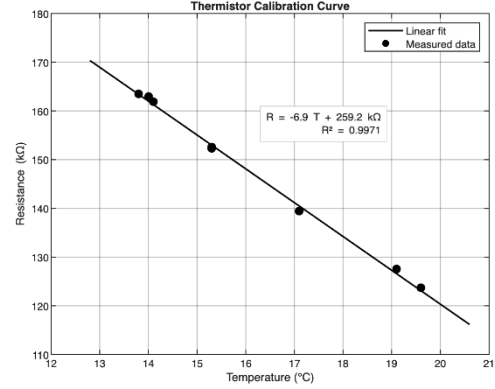


Figure 8: Calibration curve for the NTC thermistor, between sensor resistance and water temperature

Software modifications were required to process the voltage readings into temperature data in real time. The Arduino IDE code was made to first invert the voltage divider to isolate the thermistor's resistance (Eq.7)

$$R_{k\Omega} = 100 \cdot \left(\left(\frac{5}{V_{out}} \right) - 1 \right) \quad (\text{Eq.7})$$

Then, the software inverted the calibration curve to output temperature in Celsius (Eq.8)

$$T = \frac{259 - R_{k\Omega}}{6.95} \quad (\text{Eq.8})$$

To prevent errors, the code clamped the final temperature values to a minimum of 10°C and a maximum of 20°C.

However, the circuit experienced a critical hardware failure during the harbor deployment. The voltage from the thermistor spiked and railed to 3.3V as the depth increased. This indicates a physical short circuit connecting the analog pin directly to the 5V source, or a faulty ground connection across the 100 kΩ fixed resistor, likely caused by saltwater. Future iterations of this payload will require fortified epoxy waterproofing around the thermistor to ensure that the thermistor provides reliable data.

III. ROBOT DESIGN

The autonomous surface vehicle (ASV) relies on a positively buoyant primary frame that supports a negatively buoyant, winch deployed sensor payload. This design allows the vehicle to navigate the water's surface while independently collecting data vertically.

The primary frame utilizes a PVC pipe frame measuring 0.5 feet tall, 1.0 foot wide, and 1.5 feet long. To ensure stability and prevent capsizing, foam pool noodles attach to the perimeter of the frame. This distributes the buoyancy, maintaining the vehicle's center of buoyancy well above its center of mass. The vehicle navigates the surface powered by two independent side-mounted thrusters. This configuration enables the ASV to translate to the target drop zone and hold position while the winch operates.

All sensitive microelectronics, including the Teensy microcontroller and the main 12 V, 2.2 Ah battery, reside inside a sealed, waterproof enclosure mounted to the top deck of the PVC frame. The solar panel, thermistor, and pressure sensor integrate directly into the 3D printed deployable payload structure.

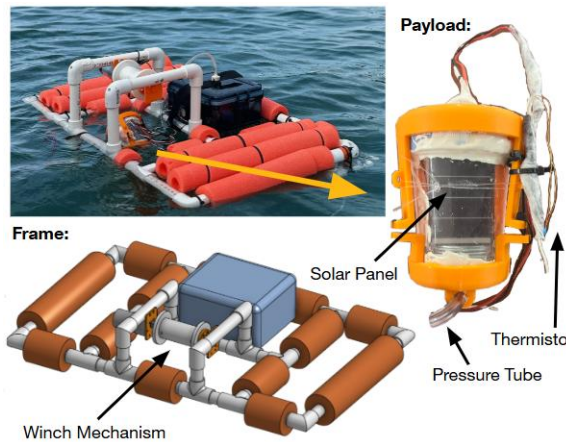


Figure 9: Mechanical drawing of the ASV and photos of the ASV and its payload

For the winch system to successfully lower the sensors, the payload must achieve negative buoyancy. The payload displaces a volume of approximately 10.5 cubic inches. The saltwater in Dana Point Harbor has a density of roughly 0.0372 lb./in³ so the submerged payload generates an upward buoyant force of 0.390 lbs. Therefore, the payload requires a total ballast exceeding 0.390lbs. to sink.

A motorized spool mechanism controls the payload's vertical positioning. The custom spool features a 0.75-inch radius. Lifting a minimum of 0.390 lb. payload requires a theoretical torque of 0.293 lb.-in. The selected winch motor provides a torque of 33 in.-lb. at 12 V and approximately 8.25 in.-lb. at 3V. At 3V, the motor operates at approximately 30 RPM. Translating this rotational velocity across the 1.5-inch diameter spool yields a controlled payload ascent and descent rate of

approximately 2.35 inches per second. This slow, speed allows the sensors adequate time to stabilize at discrete depths during the stepped descent sequence.

Saltwater is highly conductive and corrosive. While standard epoxy waterproofing proved sufficient for freshwater testing in pHake Lake, the conductive saltwater of Dana Point Harbor created problems. This environmental difference caused thermistor failure.

IV. DEPLOYMENT DETAILS

To accurately capture the nonlinear decay of the solar panel sensor without lag, the payload required stabilization periods at various depths where there would be no electrical interference from the motors. Therefore, the ASV microcontroller was programmed with a stepped descent protocol rather than a continuous drop. The winch programming lowered the payload in three second intervals with three second stabilization pauses allowing the sensors to reach equilibrium at discrete depths.

For the deployment process, initially, the 12 V battery and Teensy microcontroller were secured inside the waterproof box. Powering the Teensy with the uploaded Arduino IDE code initiated a programmed 20-second standby period. This delay provided the handling with sufficient time to securely seal the enclosure lid, place the ASV into the water, orient the vehicle's heading, and attach a tether to not lose the ASV.

Following the standby delay, the thrusters engaged for five seconds, translating the vehicle away from the launch point. The winch then executed the stepped descent sequence, lowering the sensor payload to a maximum depth of roughly 2 meters. Upon reaching maximum depth, the winch continuously reeled the payload back to the surface, and the motors ceased operation.

Upon mission completion, the ASV was manually retrieved from the water by pulling the tether. Data extraction involved removing the onboard SD card from the motherboard and processing the generated binary (LOG.BIN) and information (INF.TXT) files using a custom MATLAB script. This script decoded the raw binary bytes to generate plots for time, light intensity, temperature, and depth.

Deployments occurred in Dana Point Harbor, an urban coastal environment that has been subjected to local runoff [5]. The launch point originated from a stable docking structure. The vehicle navigated outward along a short, linear trajectory into deeper water before holding position

to initiate the vertical water data collection as seen in (Figure 10).



Figure 10: Aerial view of the Dana Point Harbor deployment location

This deployment strictly prioritized environmental, mechanical, and electrical safety. To mitigate water hazards, deployments were confined to designated low activity areas to prevent collisions with traffic. Life vests were used by all team members to ensure each other's safety as well.

Electrically, the 12 V sealed battery was isolated within a dedicated, waterproof box to prevent short circuits and potential chemical contamination of the harbor environment.

V. RESULTS

The Dana Point Harbor deployment successfully yielded synchronized data for depth, light intensity, and temperature. The extracted data was processed to evaluate the physical performance of the autonomous surface vehicle (ASV) and the optical properties of the coastal water.

The pressure sensor successfully recorded the vehicle's dive profile that matched the expected stepped descent from the winch deployment. As shown in the bottom subplot of (Figure 11), the payload descended to a maximum depth of 1.84 meters. The discrete vertical steps are visible in the depth plot perfectly matching the programmed protocol of alternating descent and stabilization phases. This confirms that the payload achieved the stable depth necessary to prevent sensor lag and ensure less interference from the motors at each measurement interval.

The solar sensor effectively captured the attenuation of sunlight through the harbor water as seen in the top plot of (Figure 11) which demonstrates the calculated light intensity decrease from an estimated 100,000 Lux at the surface down to roughly 88,000 Lux at 1.84 meters (Figure 11, top).

This steep, non-linear decline aligns closely with theoretical models of light attenuation in coastal embayments. The rapid absorption and scattering of light over a relatively shallow depth of

1.84 meters indicates a high concentration of total suspended sediments (TSS) within the harbor. Additionally, the data exhibits flatlining at the surface prior to descent confirming the theoretical expectation that measuring an unloaded open circuit voltage causes the sensor to prematurely clip at the 100,000 Lux limit in bright environments. There is a visible sudden dip in Lux when the depth is decreasing because as the payload rises, the solar panel is most likely flipped and pointed upside down, so it is not facing the sun which explains the extremely low Lux values.

The thermistor data has a critical hardware discrepancy encountered during the deployment. Initial voltage readings outside of the water exhibited standard fluctuations and matched thermometer reading as expected. However, at precisely $t=25$ seconds when the depth increases the thermistor temperature abruptly spiked and railed to 20°C .

This indicates a short circuit. Because the saltwater in Dana Point Harbor is highly conductive, it likely bypassed the sensor's epoxy waterproofing and went through the analog signal pin directly to the 5V supply line or severing the connection across the grounded $100\text{ k}\Omega$ resistor. While the exact thermal gradient of the harbor could not be fully mapped due to this issue, the failure provided evidence regarding the highly conductive marine environment.

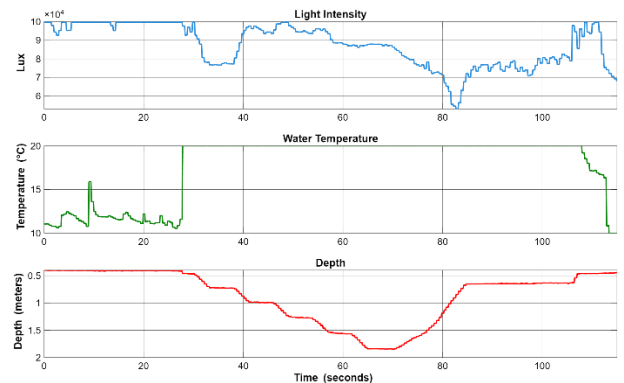


Figure 11: Calculated AUV data over a 120 second deployment.

VI. CONCLUSIONS

This project successfully demonstrated the viability of an autonomous surface vehicle equipped with a stepped-descent winch system for high-resolution environmental profiling. Deployed in Dana Point Harbor, the vehicle mapped the optical properties of the shallow water down to a depth of 1.84 meters. The collected data validated

theoretical models of coastal light attenuation, revealing a steep exponential decay in sunlight intensity that indicates significant suspended sediment within the harbor. While the mechanical systems and pressure sensors performed flawlessly, the failure of the thermistor caused hardware challenges due to the highly conductive marine environment.

The primary conclusion drawn from this deployment is that surface-tethered payloads offer a highly controlled, stable method for gathering vertical water data without the complexities of a fully submersible vehicle. Moving from simulated freshwater environments to dynamic saltwater ecosystems drastically reduces the margin of error for hardware integration.

For future iterations of this system, two critical design improvements are recommended. First, all submerged sensor leads, particularly the thermistor voltage divider, must be encapsulated with marine-grade waterproof materials rather than epoxy to prevent the saltwater short circuits experienced during this deployment. Second, the solar sensor circuit should be redesigned to include an integrated circuit that could reduce the noise. This would yield a linear response to light intensity and prevent the signal saturation and data clipping observed near the water's surface. Implementing these changes will yield a highly efficient and low-cost method capable of comprehensive coastal water quality monitoring.

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